

Calculo de valor medio

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(-\frac{5}{6}x^2 + \frac{1}{3}x - \frac{2}{7} \right) dx = \frac{1}{\pi} \int_0^{\pi} \left(-\frac{5}{6}x^2 - \frac{2}{7} \right) dx$$

q/ ser una funcion impar se elimina en el intervalo

$$a_0 = \frac{1}{\pi} \left[-\frac{5}{18}x^3 - \frac{2}{7}x \right]_0^{\pi} = -\frac{5}{18}\pi^2 - \frac{2}{7} = -3.027$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \left(-\frac{5}{6}x^2 - \frac{2}{7} \right) \cos(nx) dx$$

$$a_n = \frac{-5}{3\pi} \int_0^{\pi} x^2 \cos(nx) dx = -\frac{5}{3\pi} \left[\frac{2\pi(-1)^n}{n^2} \right] = \frac{10(-1)^n}{3n^2} = \frac{10(-1)^{n+1}}{3n^2}$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} \left(\frac{1}{3}x \right) \sin(nx) dx = \frac{2}{3\pi} \int_0^{\pi} x \sin(nx) dx$$

$$b_n = \frac{2}{3\pi} \left[-\frac{\pi(-1)^n}{n} \right] = -\frac{2(-1)^n}{3n} = \frac{2(-1)^{n+1}}{3n}$$

$$f(x) = \left(-\frac{5}{18}\pi^2 - \frac{2}{7} \right) + \sum_{n=1}^{\infty} \left[\frac{10(-1)^{n+1}}{3n^2} \cos(nx) + \frac{2(-1)^{n+1}}{3n} \sin(nx) \right]$$